ESP32-Based Sequential Three-Fluid Spray Washing System

Development Plan and Milestone 1 – Wokwi Control Simulation

Purpose: define the project concept, divide development into manageable stages, and provide concrete instructions for the first software milestone in Wokwi.

1. Project Idea and Objective

The objective is to develop an automated washing system that sequentially applies three different liquids through one common pump and one or more spray nozzles. Each liquid is stored in a separate reservoir. Electrically controlled inlet valves select which reservoir is connected to the pump. An ESP32 controls the valves, pump, timing sequence, safety interlocks and, in later stages, sensors and a user interface.

The basic process is: Fluid 1 for time t1 → controlled changeover → Fluid 2 for time t2 → controlled changeover → Fluid 3 for time t3 → cycle complete. Only one process-fluid valve may be open at a time.

2. Target System Architecture

Hydraulic concept:

```
Tank 1 – Filter – Valve 1 ─┐
Tank 2 – Filter – Valve 2 ─┼─ Manifold – Pump – Flow/Pressure sensing – Nozzle(s)
Tank 3 – Filter – Valve 3 ─┘
```

Electrical concept:

```
12 V DC ─┬─ Pump
         ├─ Valve 1
         ├─ Valve 2
         ├─ Valve 3
         └─ DC/DC converter → 5 V → ESP32

ESP32 GPIO → output drivers → pump / valves
```

The first physical prototype may use a relay for the pump, but the final system should prefer logic-level MOSFET drivers for silent operation and long switching life. Each inductive load requires flyback suppression.

3. Suggested Core Components

Subsystem	Suggested component	Notes
Controller	ESP32 DevKit	Main state-machine controller; Wi-Fi available for later stages.
Pump	Existing CrocSee 12 V diaphragm pump	Single common pump for all three liquids.
Inlet selection	3 × 12 V DC normally-closed direct-	Prefer zero differential-pressure type;

	acting solenoid valves	verify chemical compatibility.
Output drivers	Relay for early test; logic-level N-MOSFETs for final design	MOSFET solution is silent and suitable for frequent switching.
Protection	Flyback diode per inductive load	Required across pump and each solenoid coil.
Logic power	12 V → 5 V DC/DC converter	Set and verify 5.0 V before connecting ESP32.
Local filtering	470 µF electrolytic + 100 nF ceramic	Place near ESP32 power input.
Controls	START and STOP pushbuttons	STOP must override all process states.

4. Development Strategy

The project should be developed in layers. Do not start by connecting all hydraulic and electrical hardware at once. First validate the control logic, then the electrical drivers, then the hydraulic behavior, and finally add sensing and user-interface functions.

Stage 1 – Wokwi control simulation

Simulate the ESP32, three valve outputs, pump output, START and STOP inputs. Develop the state machine and safety interlocks without real hardware.

Stage 2 – Wokwi fault simulation

Add a simulated flow input and ERROR state. Verify that loss of flow, STOP and reset always force the system into a safe condition.

Stage 3 – Bench electrical prototype

Transfer the tested firmware to a real ESP32. Test the DC/DC converter, relay/MOSFET outputs and one real load at a time, initially without liquids.

Stage 4 – Hydraulic prototype with water

Install three reservoirs, three valves, manifold, pump and nozzle. Use water, optionally with food coloring, to visualize fluid transitions and measure dead volume.

Stage 5 – Process characterization

Measure priming time, flow, valve restriction and transition time between fluids. Determine whether purge/transition intervals are required.

Stage 6 – Final system functions

Add flow/pressure/level sensing, alarms, MOSFET drivers, Wi-Fi interface, recipe parameters, battery operation and enclosure.

5. Milestone 1 – Wokwi Control Simulation

Milestone 1 is complete when the complete automatic sequence can be demonstrated in Wokwi using LEDs instead of real valves and a pump, with correct timing and safe response to START, STOP and ESP32 reset.

5.1 Simulation I/O

Function	Wokwi component	ESP32 pin	Meaning
Valve 1	LED + resistor	GPIO 25	ON = Fluid 1 selected

Valve 2	LED + resistor	GPIO 26	ON = Fluid 2 selected
Valve 3	LED + resistor	GPIO 27	ON = Fluid 3 selected
Pump	LED + resistor	GPIO 33	ON = pump commanded ON
START	Pushbutton	GPIO 18	Starts an automatic cycle
STOP	Pushbutton	GPIO 19	Immediately stops the cycle

5.2 Initial Test Parameters

Use deliberately short times during simulation so that repeated testing is fast. For example: t1 = 10 s, t2 = 8 s, t3 = 15 s, and valveSwitchDelay = 500 ms. These are simulation values only; final process values will be determined experimentally.

5.3 Required State Machine

```
IDLE
  ↓ START
PREPARE_FLUID_1
  ↓
FLUID_1
  ↓ t1 elapsed
CHANGE_1_TO_2
  ↓
FLUID_2
  ↓ t2 elapsed
CHANGE_2_TO_3
  ↓
FLUID_3
  ↓ t3 elapsed
COMPLETE
  ↓
IDLE
```

The implementation should use millis()-based non-blocking timing rather than long delay() calls. This allows STOP, sensor inputs and future fault conditions to be evaluated continuously.

5.4 Changeover Logic

At each liquid change, the pump should first stop, the active valve should close, a short switching interval should elapse, the next valve should open, another short settling interval may be applied, and only then should the pump restart. The exact physical delays will later be measured with the real valves.

```
Pump OFF
→ close current valve
→ wait valveSwitchDelay
→ open next valve
→ wait valveSwitchDelay
→ Pump ON
```

5.5 Safety Rules to Implement in Simulation

- On power-up or reset: Pump OFF, Valve 1 OFF, Valve 2 OFF, Valve 3 OFF.
- Never allow two process-fluid valves to be ON simultaneously.

- In normal operation, do not allow the pump to run when all inlet valves are closed.
- STOP has priority over every state and must immediately switch the pump and all valves OFF.
- After t3, all outputs must switch OFF before returning to IDLE.
- A new START command should be accepted only when the system is in IDLE, unless a deliberate restart policy is later defined.

5.6 Serial Monitor Diagnostics

The firmware should print the current state and relevant events to the Serial Monitor. This provides a simple diagnostic interface before a web or display interface is developed.

```
SYSTEM READY
START received
STATE: FLUID_1
Pump: ON | V1: ON | V2: OFF | V3: OFF
STATE: CHANGE_1_TO_2
STATE: FLUID_2
...
STATE: COMPLETE
All outputs OFF
STATE: IDLE
```

5.7 Milestone 1 Test Procedure

Test	Procedure	Pass condition
Power-up test	Start the simulation. Verify that all four output LEDs are OFF.	Behavior matches the defined safety and sequence rules.
Normal cycle	Press START. Verify V1 + Pump, then V2 + Pump, then V3 + Pump in the correct sequence.	Behavior matches the defined safety and sequence rules.
Mutual exclusion	Observe the complete cycle and confirm that no two valve LEDs are ON simultaneously.	Behavior matches the defined safety and sequence rules.
STOP during Fluid 1	Press STOP while Fluid 1 is active. All output LEDs must turn OFF immediately.	Behavior matches the defined safety and sequence rules.
STOP during Fluid 2/3	Repeat during later stages. The response must be identical.	Behavior matches the defined safety and sequence rules.
Reset test	Reset the ESP32 during an active stage. After restart, all outputs must be OFF.	Behavior matches the defined safety and sequence rules.
Repeated cycle	Run several complete cycles to verify deterministic behavior and correct return to IDLE.	Behavior matches the defined safety and sequence rules.

5.8 Definition of Done

Milestone 1 is accepted when the simulated controller reliably performs the full t1 → t2 → t3 sequence, enforces valve mutual exclusion, reacts immediately to STOP, starts safely after reset, and provides understandable state diagnostics in the Serial Monitor.

6. Milestone 2 Preview – Simulated Flow Fault

After Milestone 1, add one simulated FLOW input in Wokwi. When the pump is commanded ON, the controller should expect a flow signal within a configurable timeout (for example 3 s). If flow is not detected, the controller should stop the pump, close all valves and enter an ERROR state. This allows the fault-handling architecture to be validated before purchasing a real flow sensor.

7. What Should Not Be Finalized Yet

Do not yet finalize the exact process times, transition/purge times, valve model, flow sensor, pressure sensor or battery system. These depend on physical measurements. In particular, the common line between manifold and nozzle creates a dead volume, so Fluid 2 will not appear at the nozzle immediately after Valve 2 opens. This must be measured during the hydraulic prototype stage.

8. Immediate Next Action

Create the Wokwi circuit with one ESP32, four LEDs, six LED resistors as required by the chosen LED wiring, and two pushbuttons. Implement the state machine and complete the Milestone 1 test procedure before connecting real valves or the pump to the ESP32.

Milestone 1 goal: prove the controller before introducing hydraulic complexity.